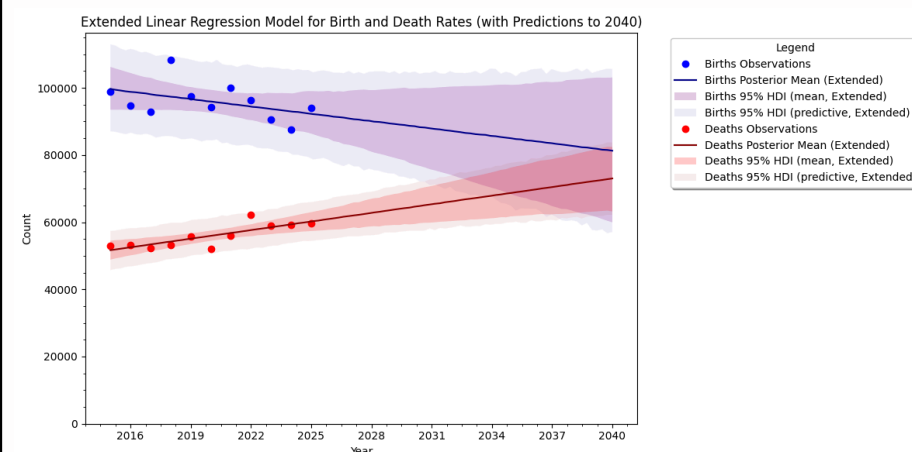


The Australian Population: Boom or Bust?

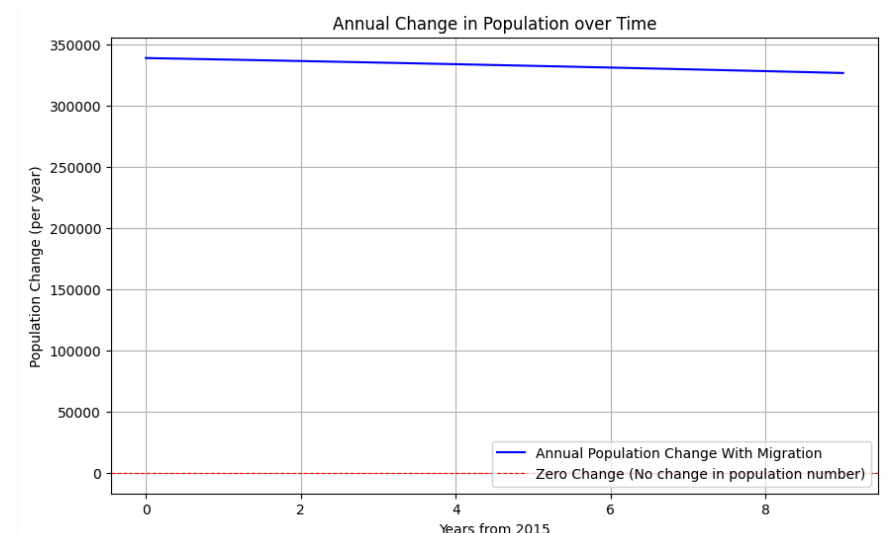
Polynomial regression of birth and death rates between 2015 and 2025 shows a downward trend in birth rates per year and an increasing trend in deaths per year. A further 5 year progression to 2030, despite the birth rate being higher than the death rate, indicate that this may not always be the case. The expectation was that as birth rate increased, death rate would increase proportionally, however as seen below, this is not the case. External factors are likely influencing the birth and death rates.



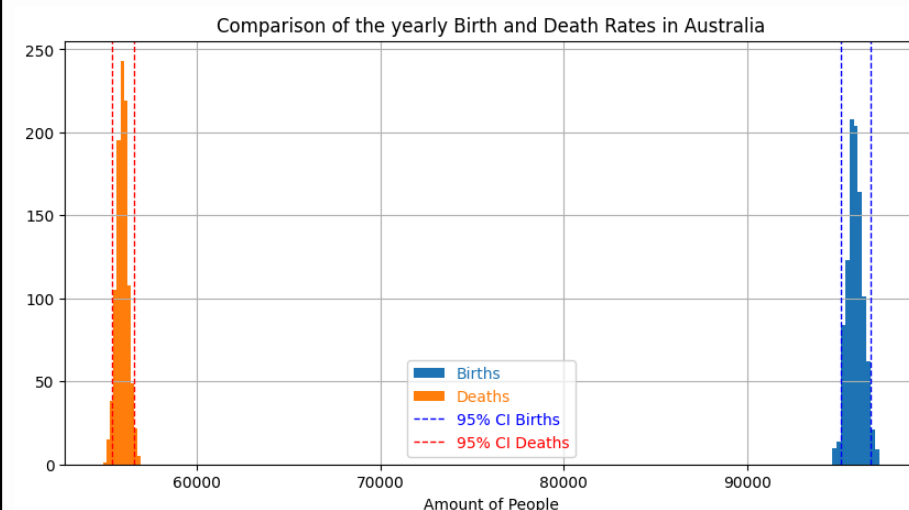
Without migration, the change in population remains above the zero-threshold line (horizontal line shown in the graph) during the simulation period showing the population doesn't decrease under the model assumptions. However, there is a clear decline illustrated suggesting the population is heading towards significant decrease should these trends continue.



With migration into Australia, the change in population remains above the zero-threshold line (horizontal line shown in the graph) during the simulation period showing the population doesn't decrease under the model assumptions. However, the increase in population mostly doesn't influence the birth rate, only the death rate as the people entering Australia were already born for the most part.



The bootstrap shows that birth rates are far exceeding death rates throughout the random sampling. This indicates that it is highly likely that there are more births than deaths during any given year.



Limitations:

Polynomial regression model

- assumes historical population trends will continue unchanged into the future
- may overfit short-term fluctuations in the data, reducing the reliability of long-term population predictions.

Bootstrapping

- is limited by the quality and size of the original dataset
- bias or inaccuracies that may have been in the original data are propagated throughout the resampling process.

Numerical model

- simplifies population change by focusing primarily on births and deaths
- may limit the accuracy of future population projections

Conclusion:

The polynomial regression analysis suggests a gradual decline in natural population growth in Australia. This trend is further supported by the numerical model when migration is not considered.

Birth rates currently remain higher than death rates however the observed trends suggest that this may not remain the case in the future.

External factors have a significant impact on Australia's overall population growth. Current migration patterns contribute substantially to population increase, indicating that despite declining birth rates, Australia's total population is still likely to experience net growth under current conditions for the foreseeable future.

